

MILESTONE – 2 REPORT

Project: MedAssist AI: Medical Symptom Analysis & Disease Prediction System

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Team: Team 4

Module: Model Training and Evaluation

Algorithms: XGBoost, Voting Classifier, Random Forest

Abstract

MedAssist AI is a machine learning based healthcare application that predicts possible diseases from symptoms entered by the user. In this milestone three machine learning algorithms were trained and evaluated. The objective was to compare their performance and identify the most suitable model for multiclass disease prediction. The project follows preprocessing, label encoding, train-test splitting, model training and evaluation using accuracy, precision, recall and F1-score.

1. Introduction

Early disease prediction helps doctors and patients take timely medical decisions. Machine learning algorithms can identify hidden relationships between symptoms and diseases by learning from historical datasets.

The MedAssist AI system uses symptom features as input and predicts the most probable disease class. Different algorithms were evaluated to improve prediction accuracy and reliability.

The implementation includes preprocessing, model training, testing and performance comparison. The best performing model is selected for future deployment into the web application.

2. Dataset Description

Property	Value
Records	189,647
Features	377
Disease Classes	587
Split	80% Train / 20% Test
Missing Values	Filled during preprocessing

3. Data Preprocessing

The dataset was loaded using Pandas. Missing values were handled, disease classes with very few samples were removed, labels were encoded using LabelEncoder, symptom columns were selected as features and the dataset was divided into training and testing sets. These preprocessing steps improved model quality and ensured consistent training.

4. Model Training

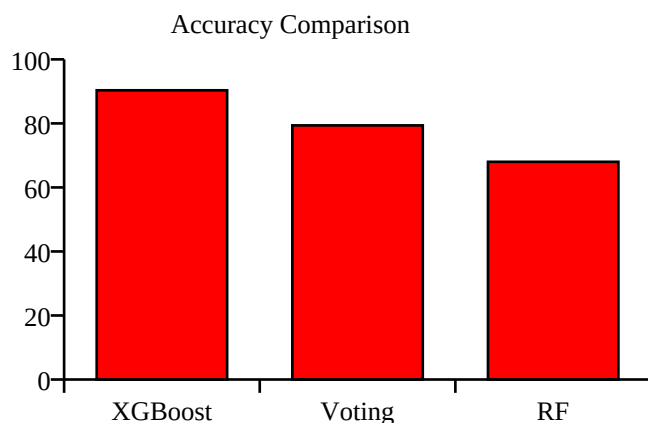
XGBoost: Gradient boosting algorithm that builds trees sequentially and minimizes previous prediction errors. It achieved the highest performance on the dataset.

Voting Classifier: Soft voting ensemble combining Decision Tree, Random Forest and Extra Trees. Final prediction is obtained by averaging class probabilities.

Random Forest: Ensemble of decision trees trained using bootstrap sampling. It provides stable predictions and reduces overfitting.

5. Experimental Results

Model	Accuracy	Precision	Recall	F1
XGBoost	90.34%	90.16%	90.34%	90.11%
Voting	79.38%	79.62%	79.38%	78.89%
Random Forest	68.00%	Not Recorded	Not Recorded	Not Recorded



6. Analysis

XGBoost outperformed the remaining models because of its boosting mechanism and efficient handling of multiclass data. Voting Classifier improved prediction stability by combining multiple models. Random Forest produced acceptable performance but lower accuracy than XGBoost.

7. Challenges

The major challenges included handling a very large multiclass dataset, memory limitations in Google Colab, increased training time and class imbalance.

8. Learning Outcomes

This milestone improved understanding of ensemble learning, preprocessing techniques, multiclass classification, performance metrics and practical implementation using Scikit-learn and XGBoost.

9. Future Scope

Future work includes hyperparameter tuning, Explainable AI integration, deployment using Flask/FastAPI and training on the complete dataset with better computational resources.

10. Conclusion

Among all evaluated models, XGBoost achieved the best overall performance and is selected as the final prediction model for the MedAssist AI system.